地电上机实验

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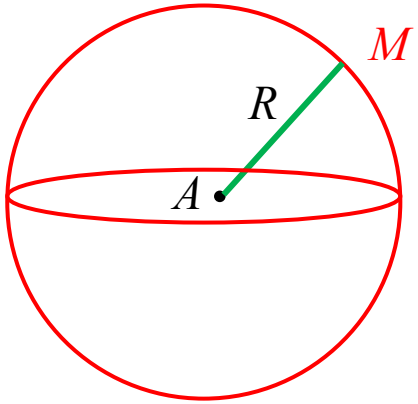
针对每种情况，画出电位分布，电流密度/电场强度的分布图

■ 全空间一个点电源的电场



$$U = -\frac{C_1}{R}$$

由于电流强度为I， 电流密度 $j = \dfrac{I}{4\pi R^2}$

$$j = \frac{1}{\rho}E = \frac{1}{\rho}\left(-\frac{\partial U}{\partial R}\right) = \frac{1}{\rho}\left(-\frac{C_1}{R^2}\right)$$

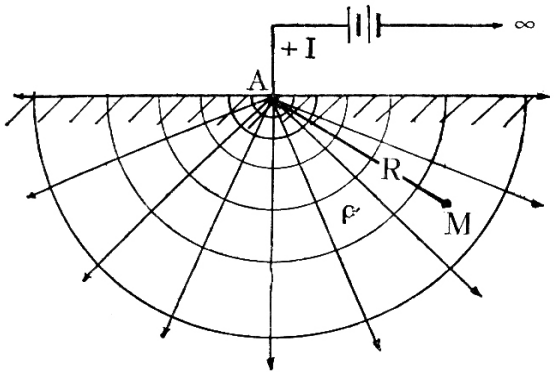
$$\boldsymbol{E} = -\nabla U$$

$$\Rightarrow \frac{1}{\rho}\left(-\frac{C_1}{R^2}\right) = \frac{I}{4\pi R^2}$$

$$\Rightarrow C_1 = -\frac{I\rho}{4\pi}$$

$$\Longrightarrow \quad U = \frac{I\rho}{4\pi R}$$

■ 半空间一个点电源的电场



电流源在地表 $U=-\dfrac{C_1}{R}$

半无限介质中，电流密度应较无限介质中大一倍，则有（镜像法或虚源法）

$$j=\frac{I}{2\pi R^2} \qquad j=\frac{1}{\rho}\left(-\frac{C_1}{R^2}\right) \qquad \Rightarrow C_1=-\frac{I\rho}{2\pi}$$

$$\Rightarrow \quad U=\frac{I\rho}{2\pi R} \qquad E=\frac{I\rho}{2\pi R^2}$$

电流源在地下

$$U = -\frac{C_1}{R}$$

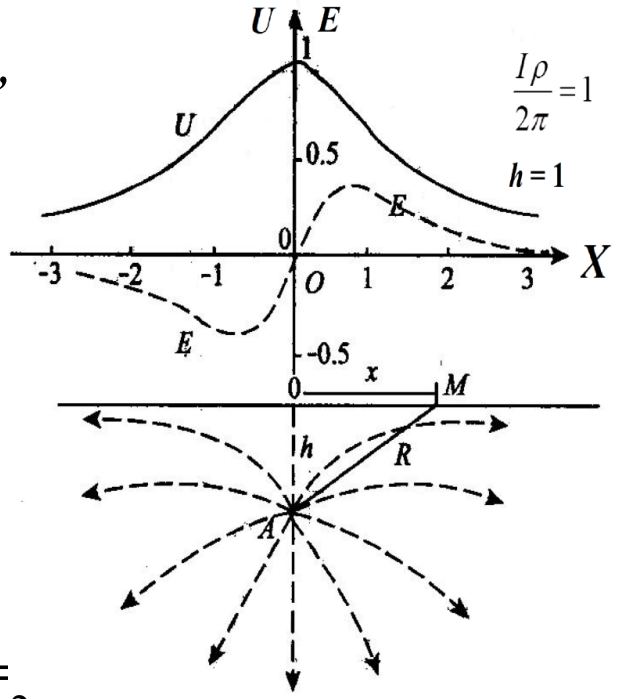
镜像法求解（关于地面对称的虚震源A'

$$U = \frac{I\rho}{4\pi}\left(\frac{1}{R} + \frac{1}{R'}\right)$$

R'为虚震源至观测点M的距离

M在地面上时，$R = R'$

$$U = \frac{I\rho}{2\pi R} = \frac{I\rho}{2\pi}\frac{1}{\sqrt{h^2 + x^2}}$$



■ 半空间一个点电源的电场

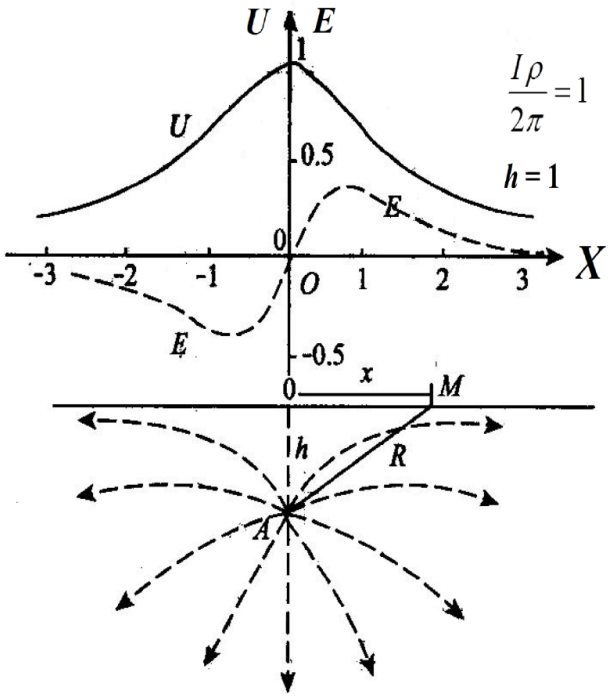
电流源在地下

沿X方向的电场强度E_x和电流密度j_x

$$U=\frac{I\rho}{2\pi}\frac{1}{\sqrt{h^2+x^2}}$$

$$E=-\nabla U\Rightarrow E_x=\frac{\partial U}{\partial x}=\frac{I\rho}{2\pi}\frac{x}{\left(h^2+x^2\right)^{3/2}}$$

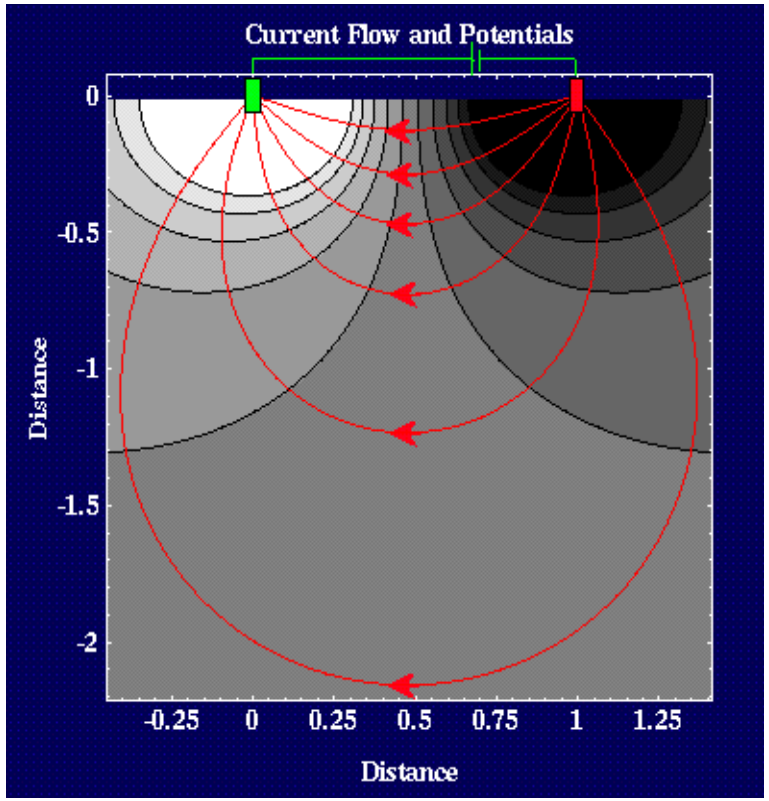
$$j=\frac{E}{\rho}\Rightarrow j_x=\frac{I}{2\pi}\frac{x}{\left(h^2+x^2\right)^{3/2}}$$



■ 半空间两个异性点电源的电场

点电源A和B相距$2L$，分别以$+I$和$-I$向地下供电，根据电场的迭加原理，便可写出A、B两电极在M点形成的电位：

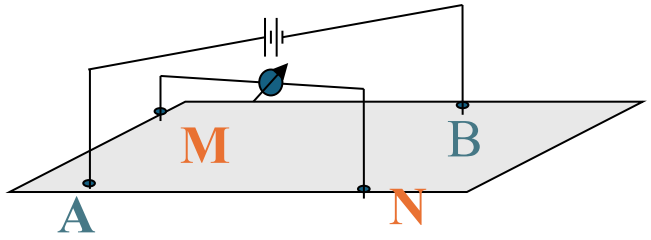
$$U=\frac{\rho I}{2\pi}\left(\frac{1}{AM}-\frac{1}{BM}\right)$$



■ 半空间地下电阻率的确定

地面水平，地下为均匀、无限、各向同性介质，将A、B两供电电极与电源相连，并向地下供入电流强度为I的电流时，则地表任意两测量电极M和N的电位:

$$\begin{cases} U_M = \dfrac{\rho I}{2\pi}\left(\dfrac{1}{AM} - \dfrac{1}{BM}\right) \\ U_N = \dfrac{\rho I}{2\pi}\left(\dfrac{1}{AN} - \dfrac{1}{BN}\right) \end{cases}$$



$$\Delta U_{MN} = U_M - U_N$$

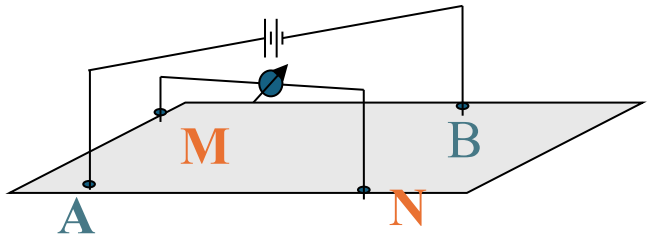
$$= \frac{\rho I}{2\pi}\left(\frac{1}{AM} - \frac{1}{BM} - \frac{1}{AN} + \frac{1}{BN}\right)$$

■ 半空间地下电阻率的确定

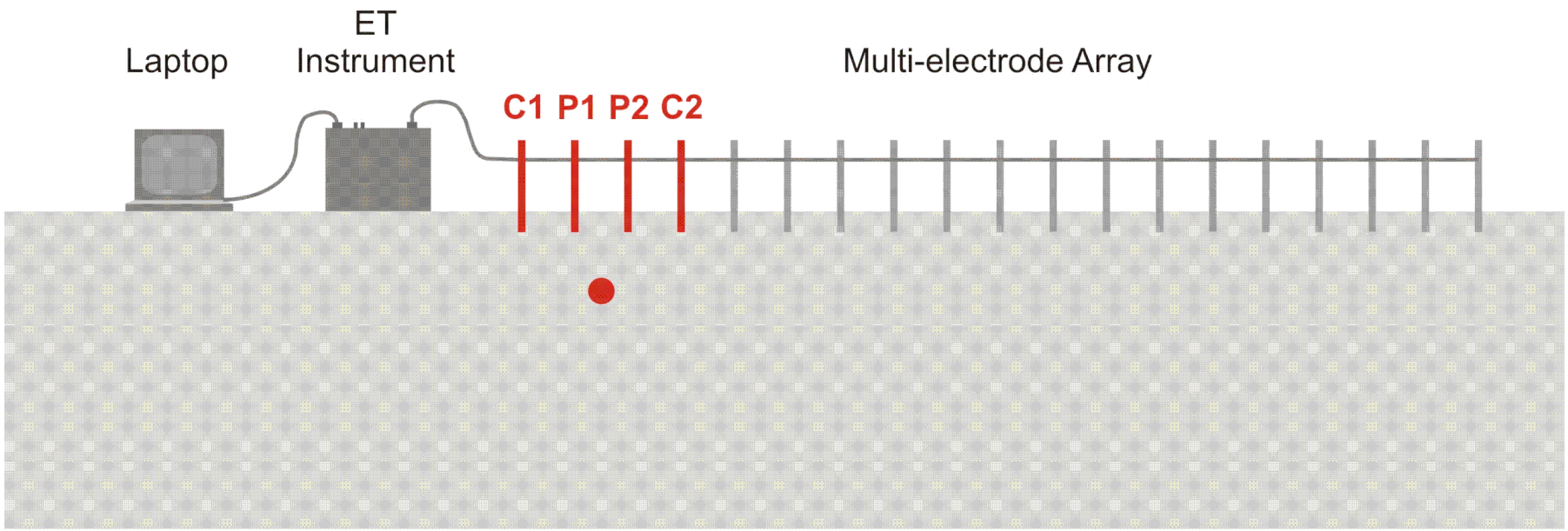
得到用点电极测量均匀大地电阻率的表达式：

$$\rho = \frac{2\pi}{\frac{1}{AM} - \frac{1}{BM} - \frac{1}{AN} + \frac{1}{BN}} \frac{\Delta U_{MN}}{I} = K \frac{\Delta U_{MN}}{I}$$

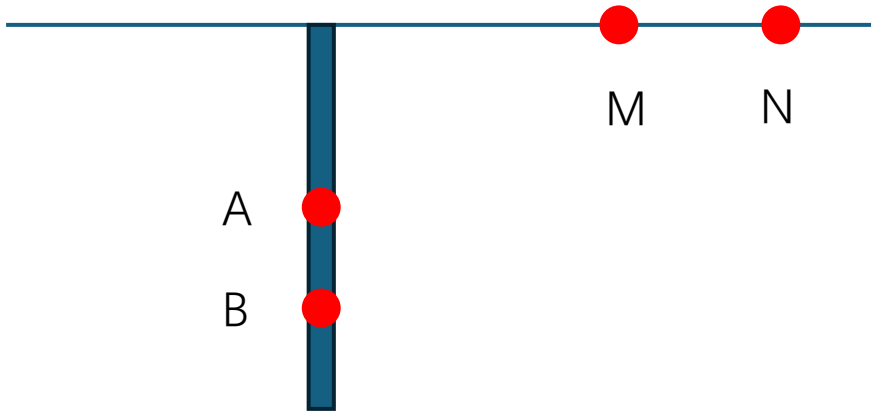
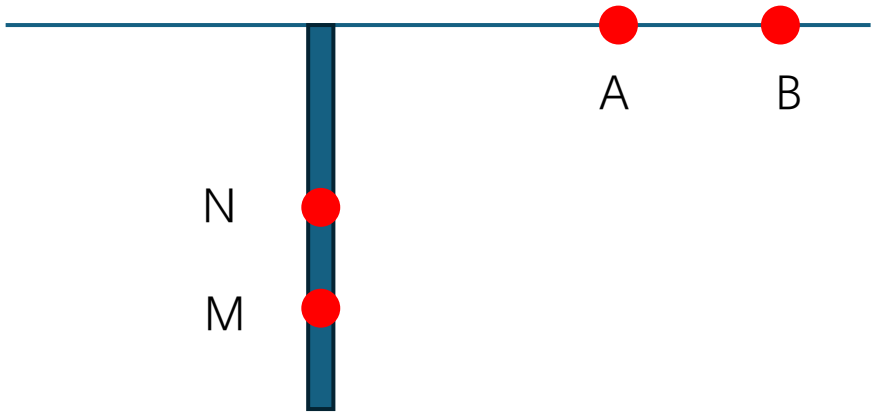
$$K = \frac{2\pi}{\frac{1}{AM} - \frac{1}{BM} - \frac{1}{AN} + \frac{1}{BN}}$$



K称为电极排列系数（装置系数），其单位为米，是一个仅与各电极间空间位置有关的量。



电极的移动、电极之间的距离变化

推导以下两种情况下MN的视电阻率公式

作图画出AB供电时，地下半空间的电流密度分布